

The Effect of Gunshot Detection Technologies on Policing Practices: An Empirical Examination of the Chicago Police Department

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Abstract: This study examines the broader consequences of gunshot detection technologies (GDTs), a controversial but increasingly common form of automated surveillance in law enforcement. While prior research has focused on GDTs' effects on firearm-related outcomes, their influence on policing behavior remains underexplored. Drawing on literature on workplace monitoring, technology adoption, and algorithmic bias, we assess how GDT deployment affects the frequency and nature of police-citizen interactions. Using district-level panel data and the phased rollout of ShotSpotter across Chicago police districts, we apply a difference-in-differences design and report five key findings. First, GDTs lead to a clear decline in citizen stops, with no evidence of differential effects by race. Second, this reduction in stops is not accompanied by increased crime. Third, crime reports are more likely to lead to an arrest, suggesting improved investigative efficiency. Fourth, citizen complaints against officers rise after GDT deployment, particularly those alleging improper search. Fifth, this increase in complaints occurs despite the decline in stops, implying a rise in complaints per interaction. These findings suggest that while GDTs may reduce the extent of routine policing, they may simultaneously intensify the enforcement actions that do occur. Our results contribute to ongoing debates about law enforcement surveillance tools, offering evidence that complicates claims that such technologies necessarily worsen or reduce bias. They also highlight the importance of institutional safeguards to ensure that gains in efficiency do not erode procedural fairness or community trust. We discuss implications for both the literature on technological monitoring and on policing technology.

Keywords: Gunshot Detection Technologies, Police-Citizen Interactions, Surveillance and Monitoring, Algorithmic Bias, Law Enforcement Efficiency

JEL Codes: K42, O33, D83

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INTRODUCTION

Organizations across the public and private sectors rely on technology to monitor workers, customers, and other stakeholders to ensure security, improve efficiency, and guide decision-making (Ball 2010, Lyon 1994, Zuboff 1988). From AI-driven performance tracking in corporate environments (Bernstein 2017, Kellogg et al. 2020), to automated surveillance in warehouses (Levy 2015, Mateescu and Nguyen 2019), to the provision of healthcare (Wachter 2012), these technologies shape employee, customer, patient, and citizen behavior, as well as organizational outcomes. One particularly prominent example of this trend is the recent and ongoing adoption of surveillance and monitoring technologies in law enforcement (Brayne 2017), the context of this study. To date, research documents that law enforcement has increasingly turned to tools such as body-worn cameras (Ariel et al. 2017, Lum et al. 2019, Zamoff et al. 2022), AI-driven decision-making systems (Richardson et al. 2019), and gunshot detection technologies (GDTs) to enhance oversight and improve resource allocation (Doucette et al. 2021).

This shift has not been without controversy. Advocates argue that monitoring technologies can enhance decision-making by reducing reliance on human discretion and enabling more efficient resource deployment. Numerous papers document these benefits across organizational contexts. For example, implementing monitoring systems has been shown to reduce employee misconduct and improve productivity in retail sales settings (Pierce et al. 2015). In healthcare, remote patient monitoring systems collect and transmit patient data to healthcare providers, facilitating timely medical decisions and improving patient outcomes (Wachter 2012). And, in policing, recent work documents that GDTs significantly improve rates of firearm and ballistic evidence recovery while also leading to reductions in shots fired (Piza et al. 2024a).

However, critics argue that these monitoring tools frequently yield unintended consequences, ranging from undesirable shifts in worker behavior (Stanton and Barnes-Farrell 1996) to resistance from employees (Bhave et al. 2020) to the creation or reinforcement of systemic biases (Yoder-Himes et al. 2022). For example, recent work in online labor markets highlights the nuanced impact monitoring technologies can have on demographic biases in job matching. On the one hand, the presence of monitoring technologies reduces employers' biases against hiring foreign workers (Liang et al. 2024). On the other hand, female workers appear less tolerant of monitoring technologies, such that the technologies induce gender-based

selection into job applications in a way that exacerbates existing gender pay gaps (Liang et al. 2018).

Similar tensions emerge in the consideration of law enforcement. As such, police departments' deployment of GDTs like ShotSpotter, which employs acoustic sensors and machine learning to detect and locate gunfire, remains intensely controversial. While GDTs have been shown to improve some first-order outcomes related to firearms-related offenses (Piza et al. 2024c), concerns abound regarding their unintended impact on systemic bias. More specifically, scholars have posited that GDTs and other law enforcement monitoring technologies may influence officer behavior in unintended ways that shape police-citizen interactions, patterns of law enforcement activity, and public perceptions of surveillance (Brayne 2017, Goold 2003, Zamoff et al. 2022). And while ShotSpotter's parent company, SoundThinking, claims that its technology "provides intelligence that allows police to coordinate a safe and efficient response, requiring fewer resources, and in a way that can enhance community trust," (SoundThinking 2024) recent evidence questions that claim (BrooklynDefenders 2024, Daniel 2024). Numerous allegations have surfaced that these technologies enable over-policing (Gee 2021), particularly in low-income communities and communities of color (Gwidt 2024, Lindenfelser 2024), that their reports are subject to manipulation by law enforcement (Feathers 2021), and that they infringe on the constitutional rights of citizens (Center 2021). These allegations have led to civil suits in Federal Court,¹ and numerous cities across the United States have terminated their ShotSpotter contracts (Ebrahimji 2024).

Yet, and importantly, while the possibility of unintended consequences is readily apparent, little empirical evidence exists to support the claim that GDTs enable systematic bias, increase police presence, or stimulate to the over-policing of citizens (minority or otherwise). Instead, extant scholarly examinations of GDTs have primarily focused on immediate, first-order effects to GDTs, such as their influence on rates of firearm recovery, violent crime, and hospital admission (Doucette et al. 2021, Goldenberg et al. 2019, Lawrence and Novak 2024, Mares and Blackburn 2021, Piza et al. 2023a, Piza et al. 2023b, Piza et al. 2024b). And what few empirical analyses *have* attempted to link racial biases to GDTs generally do so absent any

¹ 2023. *Williams V. City of Chicago*. Dist. Court, ND Illinois.

statistical means of identifying an effect (e.g. BrooklynDefenders (2024)). We address these gaps here, examining whether and to what degree the deployment of GDTs affects broader policing practices, such as the nature of police interactions with citizens, and minority citizens in particular.

As with other monitoring technologies, the relationship between GDTs and outcomes of interest (e.g., citizen stops, arrests, and any racial biases in those relationships) is unclear, *a priori*. On the one hand, as the deployment of GDTs is frequently concentrated in areas with more crime, their deployment may accelerate both legitimate and illegitimate stops of citizens. This relationship is intuitive, as the GDT likely detects gunfire that would otherwise have gone unreported. Moreover, because minorities disproportionately live in higher crime areas, which are more likely to be economically depressed (Camacho 2023, Chenoweth 2021, Patterson 1991), it stands to reason that any increase in police-initiated stops might be concentrated among minority citizens. On the other hand, because the discharge of a firearm can be detected in an automated fashion using GDTs, they may reduce police departments’ reliance on officers to proactively uncover crimes. Recognizing this, it is possible that GDT deployment may lead to a decline in the volume of investigatory stops of citizens by police officers, particularly in minority-dense neighborhoods where the technology has typically been deployed. To the extent the total volume of police interactions with citizens may decline with GDT deployment, one might even expect commensurate declines in the volumes of unjustified arrests and citizen complaints.

To identify any changes in stops and other outcomes stemming from GDT deployment, we exploit the phased rollout of ShotSpotter across Chicago Police Districts using a difference-in-differences design. Data are drawn from the Chicago Police Department’s (CPD) Investigative Stop Reports (ISRs), Crime Reports for the City of Chicago, the CPD’s Civilian Office of Police Accountability (COPA), and various other repositories. In deference to recent literature in econometrics, we employ estimators that are robust to potential biases associated with staggered treatment designs (Baker et al. 2022, Goodman-Bacon 2021), namely matrix completion (Athey et al. 2021) and Gardner et al.’s (2022) two-stage difference-in-differences estimator. The unit of analysis is the police district-month between January 2016 and December 2019.

Our analysis uncovers five notable results. First, citizen stops do not rise after ShotSpotter is

deployed in a district. To the contrary, stops decline clearly and systematically. Second, and equally important, the declines in citizen stops do not vary significantly by citizen race, suggesting that the effects are general. Third, we find no evidence that the decrease in stops is associated with increased crime, suggesting the decline in stops is not a result of ‘under-policing’. Fourth, we observe an increase in the probability that reported crimes result in arrests, suggesting that the technology does assist police in crime closure. However, fifth and finally, we find that the increase in crime closure rates may be explained by undesirable effects, namely a rise in aggressive search, as we observe a systematic increase in the volume of civilian complaints against officers, particularly complaints about improper search. This is notable because the rise in complaint volumes occurs *despite* the decline in investigatory stops. This implies a marked rise in citizen complaints per stop and suggests that declines in the extent of policing may be offset by a rise in the intensity of policing.

Numerous theoretical and practical contributions arise from these findings. First, we contribute to the growing literature on technology deployment and worker productivity (Aral et al. 2012, Bhargava and Mishra 2014, Bresnahan et al. 2002, Lu et al. 2018, Wang et al. 2024). While this literature has investigated numerous contexts, from medicine (Bhargava and Mishra 2014, Wang et al. 2024) to sales (Wu et al. 2020) and even basic science (Ding et al. 2010); such investigations of technology-enabled monitoring in policing have been limited. Moreover, what work does exist has typically focused on one form of technology: body-worn cameras (Braga et al. 2022, Zamoff et al. 2022). We build on that work by considering a technology that does not augment the execution of pre-identified tasks (Bresnahan et al. 2002, Tan and Netessine 2014, 2020, Wang et al. 2024) but instead assists workers (i.e. officers) in discovering and allocating the tasks to be performed (i.e., investigating a potential crime).

We also contribute to the literature on bias in policing (Ananthakrishnan et al. 2024, Frake and Harmon 2024, Knox and Mummolo 2020, Meuris 2023, Watson et al. 2024) and the literature on interactions between technology and racial bias (Agarwal and Sen 2022, Chan et al. 2016, Ganju et al. 2020, Rubineau and Kang 2012). While much of that literature has illustrated the disproportionate penalties suffered by social outgroups when being subject to supposedly racially blind technology, we illustrate at least one condition under which such technologies potentially reduce the burden faced by the public during the administration of

law enforcement and appear to reduce that burden equally across groups.

Our work also makes several practical contributions. First and foremost, our results inform the ongoing legal controversy surrounding the constitutionality and impacts of GDTs. While critics of these technologies have argued that they enable over-policing in minority communities (Gee 2021, Gwidt 2024, Lindenfelser 2024), our findings do not support such a claim. We observe a systematic *decline* in police-citizen interactions in GDT-affected districts, and those declines appear to manifest homogenously across racial groups, consistent with a broad decline in the extent of policing. Although we observe a marked rise in complaints, these conflicting effects raise questions about recent claims that GDTs contribute to bias in policing (BrooklynDefenders 2024, Daniel 2024), which has been used as a basis for Federal suits and as a rationale for recent city contract terminations (Ebrahimji 2024).

That GDTs reduce overall police stops while increasing rates and volumes of aggressive searches implies a complex and nuanced impact on law enforcement practices. While the reduction in discretionary stops is generally a positive development (particularly given that crime does not rise), the increase in civilian complaints suggests that officers may be shifting their approach to crime response rather than fundamentally reducing engagement with the public. This shift raises further questions about how GDTs influence policing strategies and whether additional oversight or revised protocols are needed. Furthermore, even if GDTs do not lead to racial disparities in police interactions, public skepticism and media narratives about their role in over-policing may erode community trust. Addressing these concerns requires greater transparency from law enforcement agencies regarding how these technologies are deployed and used in investigations.

Finally, and more broadly, these findings contribute to the ongoing debate about whether AI-driven law enforcement tools mitigate or reinforce bias. The fact that GDTs do not disproportionately affect minority groups challenges common criticisms of such technologies. Yet, the increase in complaints suggests that while the technology does not reinforce bias, reliance on such signals may concentrate misconduct. This suggests that other emerging policing technologies, such as predictive policing algorithms, should be examined with the potential for a similar or related dynamic in mind. Given these concerns, policymakers should carefully weigh the ethical trade-offs of using such tools and consider whether they offer a net

improvement to public safety. Additionally, courts and lawmakers evaluating suits and policy decisions related to GDTs should consider the nuanced effects these technologies have on police behavior rather than assuming they inherently and uniformly exacerbate bias.

RELATED WORK

We first provide an overview of prior scholarship discussing the role that information technologies play in organizations, both in general and those focused on monitoring. We then discuss the role of technology in policing, as well as the extant understanding of how various technologies influence police-citizen interaction. In doing so, our goal is to illustrate the relative paucity of work that has examined monitoring technology's effect on automating the tasks of workers (i.e., digital monitoring's effect beyond auditing), again both within policing and more generally. We close by discussing why the relationship between the deployment of GDTs and policing practices is theoretically ambiguous and requires further empirical investigation.

Technology, Monitoring, and Worker Productivity

Research examining the role information technology (IT) plays in shaping worker productivity has a rich history in Information Systems (IS) scholarship and has evolved along multiple paths which range from early investigations into the IT productivity paradox (Bresnahan et al. 2002, Brynjolfsson and Hitt 2000, Devaraj and Kohli 2000), to examinations of the effect of IT on worker behavior (Baker and Hubbard 2003, Duflo et al. 2012, Hubbard 2000), to recent scholarship examining the impact of digital employee monitoring.

Early analyses of the business value of IT yielded mixed evidence of the relationship between investments in IT at the firm level and firm productivity (Carr 2004). As the Nobel Prize-winning economist Robert Solow (1987, pg. 36) famously quipped: “You can see the computer age everywhere, except in productivity statistics.” This dramatically changed once scholars began to understand that productivity gains from IT were contingent on complementary organizational changes (Bresnahan et al. 2002, Brynjolfsson and Hitt 2000, Devaraj and Kohli 2000). Subsequent scholarship, in turn, began to explore how IT could enhance productivity by reshaping collaboration networks (Ding et al. 2010), how knowledge networks could efficiently foster multitasking (Aral et al. 2012), and how enterprise systems could substitute for human labor (Lu et al. 2018, Tan and Netessine 2014). Importantly, this research has continuously emphasized the role of learning curves in worker adaptation to new technologies (Bhargava and Mishra 2014), the heterogeneous

effects on workers conditional on their characteristics (Wang et al. 2024), and the importance of investing in human capital when organizing firm processes around technological capabilities (Wu et al. 2020).

In addition to the value of investing in IT, parallel streams of work have investigated how technologies shape worker behavior, influencing both productivity and organizational performance (Baker and Hubbard 2003, Duflo et al. 2012, Hubbard 2000). Studies in this domain, for example, have explored how IT can reduce illicit conduct while boosting worker productivity (Jiang et al. 2024, Liu et al. 2021, Pierce et al. 2015, Zorina et al. 2021). Where this work becomes particularly salient to our investigation is the effect of digital monitoring on employee performance. In short, this work expresses two views that frequently clash (Zorina et al. 2021); see Ravid et al. (2023) for a recent review. On the one hand, technology monitoring systems have been systemically shown to limit loss prevention in the organization and increase adherence to organizational policies. Pierce et al. (2015), for example, show how the implementation of IT monitoring systems can reduce employee theft and increase productivity, a finding echoed by Bennett et al. (2013) and Agranov and Buyalskaya (2022). Similarly, Tan and Netessine (2020) show that tabletop technology adoption at restaurants can significantly improve server performance, even if these effects are not universal. And while some scholars have argued that these effects will only manifest in the short term (Haselhuhn et al. 2012, Jiang et al. 2024), the intuition is clear: by implementing a monitoring system that ameliorates the information asymmetry between the principal and the agent via an automated auditing mechanism, monitoring systems can shift worker incentives to align with those of the principal, inducing greater worker performance while limiting free riding.

On the other hand, an equally compelling body of scholarship suggests that implementing monitoring technologies can result in conflicting psychological effects on workers (Lu et al. 2024, Patil and Bernstein 2022). Commonly referred to as “crowd-out” effects, this body of work argues that monitoring technologies can have a deleterious impact on workers by undermining trust in the employer-employee relationship (Frey 1993). Once again, this has been observed in numerous contexts, ranging from the implementation of performance minimums, which cause employees to perform to the set standard by undermining employee motivation (Deci and Ryan 2013, Falk and Kosfeld 2006), to the implementation of

location and other monitoring programs (Lu et al. 2024), which can disrupt the established routines of workers (Edmondson et al. 2001, Nelson and Winter 1982) and induce undesirable self-selection of workers into monitoring regimes (Liang et al. 2018).

Yet, despite the depth of this literature, its primary focus has been on technology-enabled monitoring for auditing compliance. To some extent, this is unsurprising because many monitoring programs are explicitly implemented to address moral hazard problems (Liu et al. 2021). However, monitoring technologies frequently take other forms and serve other purposes, namely to *aid* employees in task execution. In many contexts, employees are charged with proactively detecting and responding to external events. Consider cybersecurity, digital piracy, or social media brand sensing, where monitoring technologies regularly support employees executing their assigned tasks. In such situations, the technologies are not employed to support audit or compliance; instead, they are deployed with the primary intention of enabling workers to detect and respond to the behavior of parties external to the organization. The consequences of these technologies may be similarly impactful, whether desirable or undesirable, both for employees and external stakeholders (e.g., customers, patients, and, in our setting, citizens).

Technology and Law Enforcement

Investigations of the effect technologies have on the provision of public services have a rich tradition in the law, economics, management, and public policy literatures. In practice, scholarship in this space has primarily focused on policing because of the unique role law enforcement plays in society when administering the state's monopoly on force. Indeed, evidence suggests that IT can have immensely beneficial effects on productivity, ranging from improved emergency response times (Blanes i Vidal and Kirchmaier 2018) to enhanced crime clearance rates (Mastrobuoni 2020). And while these effects often depend on managerial and organizational adaptations (Garicano and Heaton 2010), research unambiguously indicates that improvements in policing yield reductions in crime (Di Tella and Schargrodsky 2004, Klick and Tabarrok 2005).

What role do external monitoring technologies play in this context? Unfortunately, the overwhelming body of research in this space has focused on the monitoring of individuals, and usually their specific interactions with the public, rather than broader monitoring of the public at large. For example, the rich literature on body-worn cameras (BWCs) in policing has examined everything from their effect on the use of

force (Ariel 2016, Ariel et al. 2015, Ariel et al. 2016, Braga et al. 2018, Jennings et al. 2015), to arrest rates (Ariel 2016, Grossmith et al. 2015, Katz et al. 2014, Park and Pang 2019, Wallace et al. 2018), to citizen complaints (Ariel 2016, Braga et al. 2018, Grossmith et al. 2015, Yokum et al. 2019) and de-policing (for an overview see Zamoff et al. (2022)). Indeed, recent work has even shown how technology deployments affect police data gathering (Watson et al. 2024), finding that reductions in reporting costs lead to more accurate and comprehensive data collection by law enforcement. Nonetheless, what is critical to realize is that such work has generally focused on how technologies can be used to enable monitoring for compliance, rather than augmenting worker (i.e. officer) productivity.

Research on the effect of GDTs has taken a similar path, with the extant criminology literature focusing on crime and response times rather than GDT's broader effects on police investigatory practices and behavior. Importantly, much of the work that does exist lacks a native statistical means to identify causal effects. The relative paucity of causal studies examining GDTs' broader effects may be driven in part by secrecy around the deployment of GDT microphones (Mehrotra and Scott 2024), which in turn compels researchers to rely heavily on coarse, poorly identified, approaches to evaluating their effects on outcomes like crime (Doucette et al. 2021, Mares and Blackburn 2021) and response-time (Choi et al. 2014). Only recently have researchers gained sufficient information to employ more rigorous identification strategies, but what work has been done has focused largely on first-order GDT effects, such as evidence collection, crime clearance rates (Piza et al. 2023a), emergency response time (Lawrence and Novak 2024, Piza et al. 2023b), gunshot mortality (Goldenberg et al. 2019), and crime deterrence (Connealy et al. 2024). As a result, little has been done to understand how GDTs may alter broader police behavior and interactions with citizens.

Ambiguous Predictions of GDT Deployment on Policing Practice

As discussed, GDTs represent one prominent example of a broader shift toward data-driven tools in policing aimed at improving how departments detect and respond to criminal activity. These systems use acoustic sensors and algorithmic classification to detect gunfire and relay the location of verified incidents to law enforcement in near real-time. By supplying officers with highly localized and time-sensitive information, the goal of GDTs is to reduce uncertainty in emergency response and help departments allocate resources more efficiently. However, as also discussed, the broader implications of these technologies on day-to-day policing

practices remain uncertain.

On one hand, GDTs may reduce the need for broad investigatory activity by enabling officers to respond to specific events more precisely. Instead of relying on generalized patrols or vague civilian reports, officers can use GDT alerts to focus on areas where *actual* gunfire has been detected (SoundThinking 2024). In theory, this improved targeting could lower the volume of unnecessary stops and reduce contact with individuals unconnected to criminal activity. Thus, GDTs may substitute for discretionary patrol-based enforcement by providing officers with more concrete and actionable information.

At the same time, GDTs may expand the scope of police activity by increasing the number of incidents flagged for response. Because many incidents of gunfire go unreported, GDTs may introduce a stream of alerts that would not otherwise exist. This may lead to more frequent police deployments, particularly in areas where the technology is installed. Moreover, false positives or ambiguous alerts, or those that are difficult to verify on the ground, might also prompt investigative responses, increasing the overall volume of police-citizen encounters even when no crime has occurred (Piza et al. 2023a). If this were the case, GDTs would not replace traditional enforcement, but instead add to it, expanding the number of cues that officers respond to without necessarily improving the investigatory efficiency of those responses.

A further complication arises from how officers interpret and act on GDT alerts. Because these alerts come from an automated system, they may carry an implicit authority that encourages officers to treat them as reliable signals of criminal activity. This could foster overconfidence in the technology's accuracy and lead to more aggressive investigatory tactics, especially when alerts are treated as grounds for search or detention. As a result, GDTs might be used to justify pretextual stops, encounters in which an officer uses the alert as a nominal justification for investigative activity that is, in practice, driven by other factors, such as an underlying racial bias or general suspicion. This dynamic raises the possibility that GDTs, even when accurate, could be leveraged to legitimize stops or searches that would otherwise lack clear legal justification, particularly in communities that have historically experienced disproportionate police scrutiny.

More specifically, the effect of GDTs on investigatory practices may vary across racial groups, even when the alerts themselves are race-neutral, because GDTs are overwhelmingly deployed in high-crime

neighborhoods, which are, in many cities, disproportionately home to minority residents. As a result, many citizen encounters stemming from GDT alerts will likely involve minority citizens simply by virtue of where the alerts are generated. This introduces a structural pathway through which race-neutral technology may contribute to racially unequal enforcement. Thus, should officers use GDT alerts, regardless of their specificity or validity, as *post hoc* justification for pretextual stops, the volume of stops targeting minority citizens could increase, even in the absence of any change in underlying behavior. In this way, the technology may serve not only as a detection tool but also as a legitimating device for aggressive policing practices in minority communities. As a result, GDT deployment could amplify existing disparities in stop rates, not because the GDT directs officers to specific individuals, but because it provides a flexible rationale for investigative behavior in neighborhoods that are already overrepresented in police activity.

Yet, it is also plausible that GDTs might reduce racial disparities in policing. Because alerts are generated based solely on detected locations and timing, and are decoupled from officer judgment or citizen reports, they may reduce the influence of race-based heuristics in determining when and where to investigate. This mechanism mirrors findings from “veil of darkness” studies, which show that racial disparities in traffic stops decline when officers cannot perceive the race of the driver at the time of the stop decision (Grogger and Ridgeway 2006). Likewise, research on automated traffic enforcement suggests that race-blind triggers can yield more equitable outcomes (Xu et al. 2024). If GDTs act as an objective trigger for police response, they may shift enforcement decisions away from subjective bias, reducing disparities in investigatory stops across racial groups, even in neighborhoods where racial disparities have historically been most pronounced.

Taken together, these competing mechanisms paint a murky picture of the potential effect of GDT deployment on police behavior. On one hand, GDTs may intensify pretextual or aggressive policing of citizens by legitimizing discretionary stops, either in neighborhoods already subject to heightened enforcement or in general. On the other hand, the standardization of enforcement cues may diminish the proactive role law enforcement plays in initiating contact (either broadly or among specific groups). Which effect dominates is therefore an empirical question, and one that depends not only on the design of the technology, but also on how officers leverage it in practice.

As such, the broader impact of GDTs hinges on how they are integrated into policing strategy and departmental norms. If GDTs are used to support focused, evidence-based enforcement with clear protocols governing officer conduct, the technology may enhance both efficiency and equity in policing. But if GDT alerts are used to rationalize aggressive tactics that exacerbate enforcement outcomes in minority communities, the technology may produce or intensify the same disparities that technology-based interventions are often presumed to address. These competing possibilities highlight the importance of empirical evaluation and policy oversight in shaping the role of monitoring in contemporary law enforcement.

METHODS

Data Sources and Variables

We analyze four categories of outcomes: citizen stops, crime statistics, arrest patterns, and civilian complaints, using administrative data from the Chicago Police Department (CPD) and affiliated sources. The sample is operationalized at the police district-month level between January 2016 to December 2019. Citizen stop data is drawn from Investigative Stop Reports (ISRs), from which we aggregate counts of pedestrian stops split by suspect race (White and non-White). Crime reports with arrest indicators come from the CPD's public portal, with closure rates defined as the proportion of crimes indicating arrest. CPD arrest data with charging statutes were obtained through Freedom of Information Act (FOIA) requests. Arrest counts are aggregated to the district-month along with firearm-specific arrests (discharge offenses and unlawful use of weapon). These offenses are identified using FBI offense codes and the statute descriptions associated with each arrest. Counts of civilian complaints against officers, categorized by type (e.g., improper search), come from the Civilian Office of Police Accountability (COPA). We also collected data on monthly district staffing headcounts, excluding recruits, which were obtained through Freedom of Information Act (FOIA) requests. ShotSpotter deployment dates across districts are drawn from Connealy et al. (2024), and are validated against CPD's ShotSpotter Alerts data, as described below. Count outcomes variables (viz., stops, crimes, arrests, complaints) are log transformed as $\ln(y_{it}+1)$. Closure rates remain in linear form. All variables are further defined in the appendix. Summary statistics are in Table 1.

-- INSERT TABLE 1 HERE --

ShotSpotter Deployment

Our main variable of interest, ShotSpotter_{it} is set to 1 in police district i for all month-year t after the

technology has been deployed in that district. To confirm the validity of the schedule from Connealy et al. (2024), we further regress (1) the number of calls initiated by ShotSpotter in district i and (2) the number of rounds detected by GDT in district i on the ShotSpotter_{it} indicator (see Appendix Table S1). As expected, deployment correlates strongly with increasing ShotSpotter-initiated calls ($p < 0.01$) and number of rounds detected ($p < 0.01$). Figure 1 illustrates the geographical rollout of ShotSpotter across Chicago police districts and Figure 2 depicts the timing of rollout for GDT-receiving districts.

-- INSERT FIGURE 1 HERE --

-- INSERT FIGURE 2 HERE --

Empirical Strategy

To determine the effect of ShotSpotter deployment on our various outcomes we leverage a staggered difference-in-differences (DiD) design. The unit of analysis is the police district i in month-year t . The treatment indicator ShotSpotter_{it} equals 1 post-deployment. To illustrate our empirical strategy, consider a baseline difference-in-differences specification incorporating district and time fixed effects below.

$$\text{Outcome}_{it} = \beta \text{ShotSpotter}_{it} + \theta_t + \eta_i + \varepsilon_{it} \quad (1)$$

Under standard identifying assumptions (i.e., parallel trends, no anticipation, SUTVA, and treatment effect homogeneity), β estimates our parameter of interest, i.e., the average treatment effect of ShotSpotter deployment on the treated (ATT) for chosen outcomes. At this point, it is important to note that GDT deployment is not assigned at random and that our goal is not to estimate an average treatment effect (ATE). GDT deployment is not random across geography because city officials deploy GDT in areas where gun crime is more prevalent. This does not undermine our empirical strategy for two reasons. First, although GDT deployment may be a function of gun crime, our main outcomes of interest are police investigations, not crime itself. To the extent that crime drives investigations, influence from time-invariant differences in crime across districts will be absorbed by district fixed effects. It should also be noted that there is a risk that the trajectory of crime drives GDT deployment in a manner that confounds identification. We address the potential time-variant selection issues in the Robustness section (see Figures 5 and 6). Second, although GDT deployment is non-random, our primary empirical objective is to understand what has happened in districts that received GDT (ShotSpotter). Thus, the relevant parameter of interest is the ATT, not the ATE. This

distinction is critical as our goal is to inform city managers and policymakers of the expected effect in areas that deploy similar monitoring technologies in the future, and any such generalization will likely align with jurisdictions having similar characteristics to those that were treated in Chicago.

In lieu of a standard two-way fixed effect (TWFE) estimation using an OLS, we follow recent advances in the econometrics literature addressing phased treatment DIDs (Athey et al. 2021, Gardner 2022, Goodman-Bacon 2021). This approach relaxes the assumptions of a TWFE model to address potential bias stemming from staggered adoption. Thus, rather than estimating the TWFE model with linear fixed effects regressions, we implement two contemporary approaches: matrix completion via the R package ‘fect’ (Athey et al. 2021) and Gardner's (2022) two-stage DiD estimator (DID2S). Robust standard errors are clustered at the police district level.

RESULTS

Investigatory Stops and Potential Racial Disparities

In Table 2, we report estimates of the relationship between ShotSpotter deployment and the volume of investigative $Stops_{it}$. Results indicate a significant decline in citizen stops following ShotSpotter deployment. Specifically, districts equipped with the technology experienced a 17.6% reduction in total investigatory stops ($\beta = -0.194$, $p < 0.05$), relative to untreated districts, with effects persisting throughout the post-deployment period (Figure 3). Next, in deference to claims that ShotSpotter deployments result in over-policing of minority communities (e.g., *Williams v. City of Chicago*, Gwidt 2024), we decompose stops into those involving White versus non-White suspects. Contrary to claims of GDT enabling racially targeted over-policing, stops decline across both racial groups, with no significant difference across White and non-White citizens (i.e., we cannot reject the null of no difference in this effect). White stops decreased by 20.8% ($\beta = -0.234$, $p < 0.05$), while non-White stops fell by 16.2% ($\beta = -0.177$, $p < 0.1$). These results undermine the notion that ShotSpotter is used to justify increased police presence in a community.

-- INSERT TABLE 2 HERE --

-- INSERT FIGURE 3 HERE --

As discussed, Figure 3 plots the dynamic effects for each outcome. In it, we observe a clear and consistent decline in stop reports after ShotSpotter is deployed, and no significant changes are observed in

the months leading up to ShotSpotter’s arrival. These results suggest the effect is persistent. Further, we observe no evidence that the parallel trends assumption is violated (Angrist and Pischke 2009), suggesting that the model is properly specified. GDT thus appear to change police practices, resulting in a persistent decrease in presumably unnecessary citizen stops. This raises another potential threat to identification. It is possible, for example, that GDT are viewed as a substitute for human patrols by supervisors, leading to a reallocation of patrol units away from GDT-treated districts (i.e. supervisors might assign fewer officers to GDT treated districts, thereby permitting the technology to pick up the slack). This might plausibly yield the observed decrease in investigatory stops due to reduced patrols. We find no evidence to support this potential mechanism, as there is no evidence that staffing levels are reduced for districts receiving the technology. In Column 4, we regress *District Staffing_{it}* on GDT rollout and find an insignificant effect on staffing with a magnitude near zero, which we interpret as an economic zero. Dynamic treatment effect plots for district staffing levels are in Appendix Figure S1, where we observe no pre- or post-treatment divergence from the baseline.

Crime Dynamics and Investigative Efficiency

We next turn to the effect of ShotSpotter’s deployment on district crime reports and crime closure rates.

Results are in Table 3. In Column 1, we observe a negative, though statistically insignificant, effect on the total number of reported crimes. Given this result, we have no evidence that the decline in police investigative stops translates to a rise in crime. We observe no evidence that ShotSpotter affected the total volume of gun crime in Column 2, and thus the decrease in investigatory stops cannot be explained by a change in the extent of gun crime. In Column 3, we observe a statistically significant rise in the probability that reported crimes result in an arrest, equal to approximately 1.3pp. Thus, while crime volumes do not appear to rise, closure rates do, which is desirable. Finally, we narrow this result and examine the effect of ShotSpotter deployment on arrests for crimes involving illegal gun discharge and unlawful use of weapons (UUW). Results are in Columns 4 and 5 of Table 3 and indicate a significant rise in such arrests, a result consistent with the findings of Connealy et al. (2024). Specifically, we find that ShotSpotter increases arrest volumes for both types of crimes (gun discharge arrests up 11% and UUW arrests up 13.6%). Thus, and encouragingly, despite the drop in investigatory stops, police appear to be solving a greater percentage of

reported crimes, at least as proxied by arrest rates. Dynamic treatment effects for these outcomes are in Figure S2 of the appendix, which once again indicate that the effect is persistent with no significant pre-treatment effects manifesting.

-- INSERT TABLE 3 HERE --

Complaints and Investigative Intensity

Thus far, our findings are consistent with the claims put forward by ShotSpotter’s parent company; on average, GDTs appear to enhance the efficiency of policing by decreasing the volume of police investigatory stops targeting civilians while not adversely affecting the rate of crime or the crime closure rate. However, it is possible that this apparent efficiency manifests only at the extensive margin (i.e., the volume of investigatory stops), with different consequences manifesting at the intensive margin (i.e., a possible increase in the aggressiveness of law enforcement when stops do occur). Indeed, allegations against ShotSpotter and its host cities speak specifically to the idea that ShotSpotter provides a pretext for search and harassment upon officers’ arrival at the scene following a ShotSpotter call (Gee 2021). To test this possibility, we next estimate the effect of ShotSpotter deployment on the volume of civilian complaints against officers. In doing so, we consider both complaints in general as well as specific types of complaints.

Results are in Table 4. In Column 1, we observe a large (32%) and statistically significant increase in civilian complaints after ShotSpotter’s deployment. To understand the nature of those complaints, we decompose them into two specific types: complaints regarding officers brandishing firearms and complaints of improper search. We might anticipate a rise in officers brandishing firearms as a protective measure, notably if they anticipate an armed suspect, having arrived at the scene of a ShotSpotter alert. Further, based on the allegations raised in the media and anecdotal evidence noted in prior legal literature, we might anticipate a rise in aggressive search tactics. Results suggest that while complaints about officers brandishing firearms appear to decline (Column 2), complaints about improper search (Column 3) rise sharply (68.4%). As before, we visualize these effects over time to assess the parallel trends assumption (Figure 4). The dynamic estimates again indicate i) an increase in the volume of total complaints, ii) an apparent decrease in complaints involving officers’ brandishing firearms, iii) a dramatic increase in the volume of improper search complaints, and iv) no demonstrable violation of the parallel trends assumption before ShotSpotter

deployment for any of these outcomes. Taken in sum, these findings suggest that while individual stops of citizens are falling once GDTs are deployed, there is an increase in complaints regarding improper search (albeit without a concomitant increase in officers brandishing firearms).

-- INSERT TABLE 4 HERE --

-- INSERT FIGURE 4 HERE --

ROBUSTNESS

Dynamic Selection

An initial concern with our empirical design is that GDTs are selectively deployed to areas where gun crime is known to occur most. And while we observe no significant deviations in the effect pre-treatment, it is worthwhile to consider how selective deployment might influence our identification of the effect despite the fact that our outcome of interest is police investigations as opposed to gun crime in particular.

Several initial considerations cut away at this concern preemptively. Specifically, any time-invariant differences in crime that drive police investigations should be absorbed by the district fixed effects. Still, the potential for dynamic selection into treatment persists (i.e., it may be that districts experiencing a rise in crime over a lengthier period may be chosen for earlier ShotSpotter deployment, and that same rise in crime might also affect the extent of police investigations). To rule out the possibility that our estimates are driven by this sort of dynamic self-selection into treatment, we test whether recent changes in crime within a district are meaningfully associated with the timing of ShotSpotter deployment. Specifically, we estimate a pair of linear probability models (LPMs), wherein we regress the indicator of ShotSpotter deployment onto lagged monthly measures of crime (either in levels or logs) in the district up to one year in advance. Intuitively, such an approach is appealing because it allows us to capture whether a nascent spike in crime is the basis for deployment. Results are depicted graphically in Figure 5. As can be seen, regardless of whether we consider raw or logged measures of crime, we observe no evidence of an association. This suggests, all else equal, that the original estimation is properly specified, and that city managers are not dynamically reacting to changes in crime when deploying GDTs.

-- INSERT FIGURE 5 HERE --

Leave One Out Analysis

Another concern is that the observed results might be an artifact stemming from idiosyncratic changes that

are attributable to a small number of treated districts rather than being a broad result. For example, it is possible that large changes in the number of stops occurred in a single district, and that district was unique in its reaction. Still, the effect size was sufficiently large to observe significance more broadly. Were this the case, our findings' generalizability would be questionable. To address this, we conduct a leave-one-out style analysis. In doing so, we repeatedly estimate our two key models (effects on stops and complaints), systematically omitting each police district from the sample, one by one. This exercise aimed to gauge our estimates' sensitivity to any single district's exclusion. If consistent results manifest, we can be reasonably sure that no one district is driving the lion's share of the effect. Results are in Figure 6, including point estimates and 90% confidence intervals for both *Stops* and *Complaints*. As the figure suggests, the effects remain regardless of which district we omit. These results indicate that no single police district drives our findings. Instead, the estimates reflect a broad and consistent pattern driven by the rollout of GDTs.

-- INSERT FIGURE 6 HERE --

Motor Vehicle Stops

In our main estimations, our primary outcome, *Stops_{it}*, captures pedestrian stops of civilians by law enforcement. This approach is intuitive as officers will likely focus investigative efforts towards citizens on foot near the locus of a ShotSpotter alert. However, the concern with such an approach is that perpetrators might react to the deployment of ShotSpotter, and the heightened ability of law enforcement to react to the unlawful discharge of a firearm and pivot towards the use of motor vehicles. If this were the case, we might expect to see a decline in the number of pedestrian stops but an increase in the number of motor vehicle stops. This would be problematic because it would simply suggest a substitutive effect, wherein the net number of citizen stops does not actually decline.

To safeguard against this possibility, we replicate our estimations using an alternative dependent variable, *TrafficStops_{it}* (i.e. district level stops of motor vehicles in the month-year). Dynamic treatment effects for this outcome are in Figure S3 of the appendix, where we again observe a large and statistically significant negative effect of ShotSpotter deployment, as is consistent with our earlier results. All else equal, the consistent decrease across all measures of investigatory stops suggests that there is not a simple substitution in the venue of where firearms are discharged or the mode of transportation for those discharging a firearm.

Alternative Estimators and Outcomes

As noted previously, the primary estimator we employ is the matrix completion estimator developed by Athey et al. (2021). To ensure our findings are not an artifact of the selected estimator, we next replicate each of our estimates using the Gardner (2022) two-stage difference-in-differences estimator, which is also robust to potential sources of bias associated with staggered treatments. Results are in the statistical appendix (Tables S2 - S4) and remain consistent in effect sizes and inference to our main results. Further, dynamic treatment effect plots (Figures S4-S6) indicate consistent results in relative time as the matrix completion estimation. Taken in sum, this suggests our main findings are not an artifact of the selected estimator.

DISCUSSION

Reports and concerns of racially biased policing have become pervasive in the United States and ample empirical evidence exists to support them (Gelman et al. 2007, Watson et al. 2024). Still, it is critical that any emergent claims be subjected to rigorous analysis to inform policy responses. This is true for at least two reasons. First, to the extent that violations of the Fourth and Fourteen Amendment rights of citizens carry heavy penalties for both individual officers and the cities they serve (see *Floyd V. City of New York*. (2013)), unfounded accusations should not be unfairly raised against them and their Due Process rights should not be violated. Second, to the degree that the resources to correct issues of biased policing are not infinite, empirical validation of such claims is critical to ensure the appropriate allocation of investigative and judicial resources.

Against this backdrop, we evaluated the impact of GDTs, namely ShotSpotter, on various policing measures. As discussed, ShotSpotter has become a lightning rod in recent years, with claims that the technology is flawed, ineffective at reducing crime, and facilitates over-policing (Center 2021, Ebrahimji 2024, Gee 2021, Lindenfelser 2024). These claims are plausible, as GDTs are known to be deployed most heavily in minority communities and high-crime areas, raising the prospect of disparate impacts and unfair treatment. However, GDTs may also yield tangible benefits, and some evidence of this has been reported in the domain of gun crime (Connealy et al. 2024). Past work notes that GDTs may lead to more efficient police investigations by supporting rapid response times, guiding spatially targeted canvassing and investigation, and providing officers with additional information. This could, in turn, lessen the burden on police patrols.

Weighing these two possibilities, we provide evidence that both effects are present. On the one hand,

GDTs appear to increase the efficiency of police investigations, markedly reducing the volume of investigatory stops, yielding improvements in crime closure rates, and producing no observable increase in total crime volume. On the other hand, our findings suggest that citizen stops that occur after ShotSpotter is deployed may be more aggressive, producing a rise in complaints against officers for improper search. The net result is, at best, murky, as the implications for social welfare are challenging to determine. This is because such a calculation would require weighing the costs of complaints against the benefits of crime clearance. Regardless, this is precisely the calculation that appears to face legislators and city managers across the country. Still, the rise in complaints may be addressable through policy intervention. To the extent complaints might be reduced through policy enforcement and training, the upside of GDTs might be better realized.

For the literature on monitoring technologies, our work bears important theoretical implications for how such technologies may lead to unintended consequences for interactions between organizations and external stakeholders. Specifically, our findings suggest that while monitoring tools can improve efficiency by reducing the volume of routine investigative actions, they may also alter the nature of remaining interactions in ways that introduce new risks. This aligns with prior work on surveillance and worker monitoring, which has shown that digital oversight can both enhance productivity and generate resistance or unintended behavioral shifts (Bernstein 2017). In context, while GDTs appear to decrease the frequency of police-citizen encounters, they also seem to escalate the intensity of investigative actions when they do occur, leading to more aggressive search behaviors and subsequent complaints. These findings highlight the broader organizational challenges associated with integrating automated monitoring technologies into decision-making processes, particularly in contexts involving discretionary human judgment and high-stakes interactions.

Our work also makes an important contribution to the literature on GDTs in law enforcement and to the literature on monitoring technologies more broadly. While the bulk of prior work has examined the efficacy of such technologies for identifying and resolving first-order outcomes of interest (i.e., those activities the technology was explicitly designed to monitor for and detect), our work is the first we are aware of to explore the consequences of externally focused monitoring technologies for a broad swath of organizational outcomes. Our study therefore extends research on the impact of law enforcement technologies by

illustrating how digital monitoring tools do not necessarily lead to over-policing, at least in terms of the extent of police interaction with citizens. Nonetheless, our work also highlights that such technologies may reshape policing activity in a more nuanced way. Specifically, our findings suggest that GDTs act as a substitute for in-person patrols, reducing the volume of citizen stops while simultaneously increasing the investigative intensity of citizen encounters when they do occur. Our work has obvious practical relevance, given the ongoing debate in major cities around the United States regarding the costs and benefits of ShotSpotter. Moreover, it highlights the need for law enforcement agencies to implement clear policies on how GDT-generated alerts should be used. This should ensure that the technology enhances efficiency without leading to unnecessary escalations in police-citizen encounters.

This study is, of course, subject to limitations, which create rich opportunities for future research. First, we leverage a quasi-experimental design tied to the phased deployment of ShotSpotter across Chicago police districts. Chicago likely has unique policing policies, practices, and populations that may influence the effects we observe. Thus, these effects might not generalize well to all cities. Future work should employ similar research designs to understand the downstream implication of GDT deployment in other jurisdictions, and how contextual differences across those jurisdictions shape the effects.

Second, our work relies on administrative data (e.g., recorded stops, arrests, complaints). Inasmuch as such data may not capture the full spectrum of police-citizen interactions or community perceptions (Watson et al. 2024), our results should be interpreted with this in mind. Furthermore, while our difference-in-differences design identifies changes in investigations and outcomes due to GDT deployment, pinpointing the precise behavioral mechanisms that are driving efficiency gains remains challenging. Future work should delve into these questions, either quantitatively or qualitatively, to better understand these mechanisms and how police-citizen interactions have changed. Third, our measure of crime clearance is focused on arrests; we do not observe post-arrest outcomes. Nor can we tie an individual GDT alert to an individual arrest. This is an unfortunate reality of the current state of police administrative data and the fact that post arrest outcomes are challenging to examine empirically. In short, this challenge centers on the fact that “wins” and “losses” are hard to define, and outcomes run the gamut from conviction, to dismissal, to adjournment in contemplation

of dismissal, to plea deals, to false confessions, to an unwillingness of prosecutors to press charges for any number of reasons within their discretion. Accordingly, our proxies for policing efficiency are, in some sense, noisy. Future work would do well to connect arrests to trial outcomes to gain a more holistic understanding of outcomes.

Data and materials availability: Data and code are here: <https://osf.io/tnwv8/>

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FIGURES

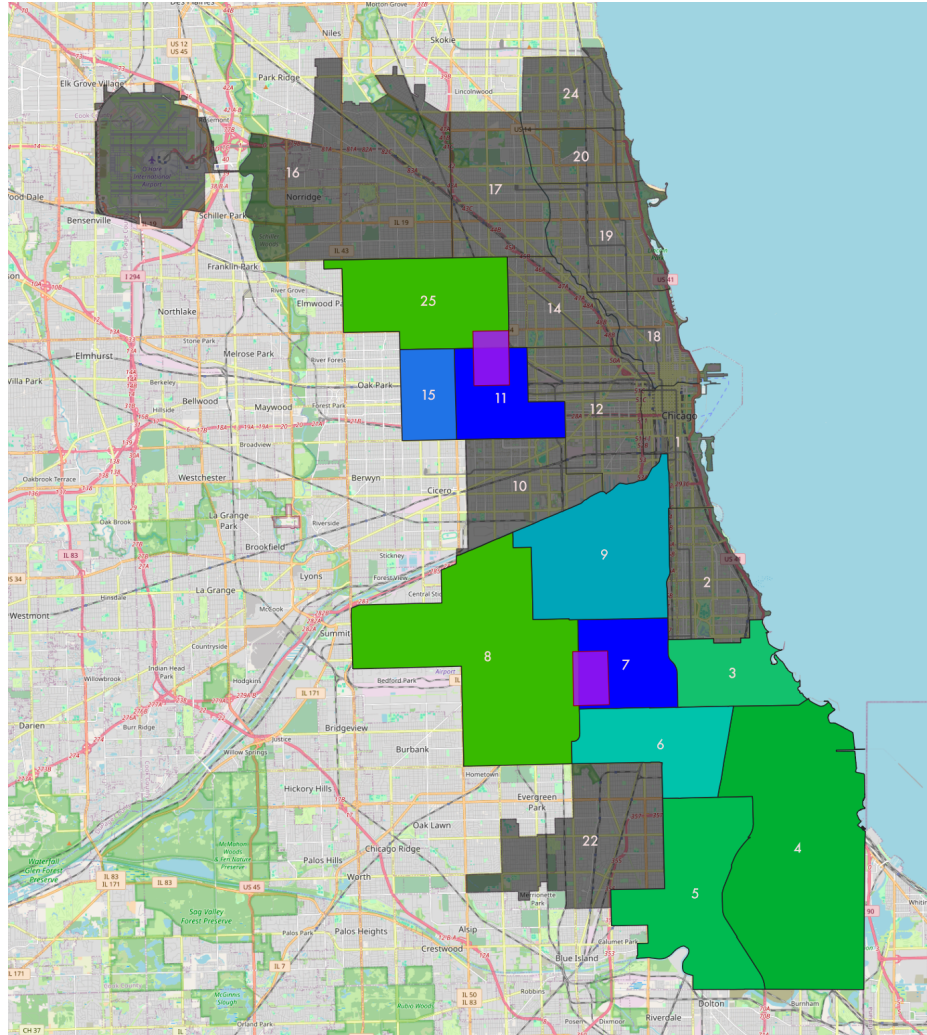


Figure 1. ShotSpotter deployment schedule.

Blue to green reflects earlier to later deployment, and gray implies no deployment. The areas in purple reflect a limited Phase 1 trial that was begun much earlier, in 2012. In each case, those limited deployments were primarily contained within a single district (7 and 11). Omitting districts 7 and 11 from our analyses yields consistent results.

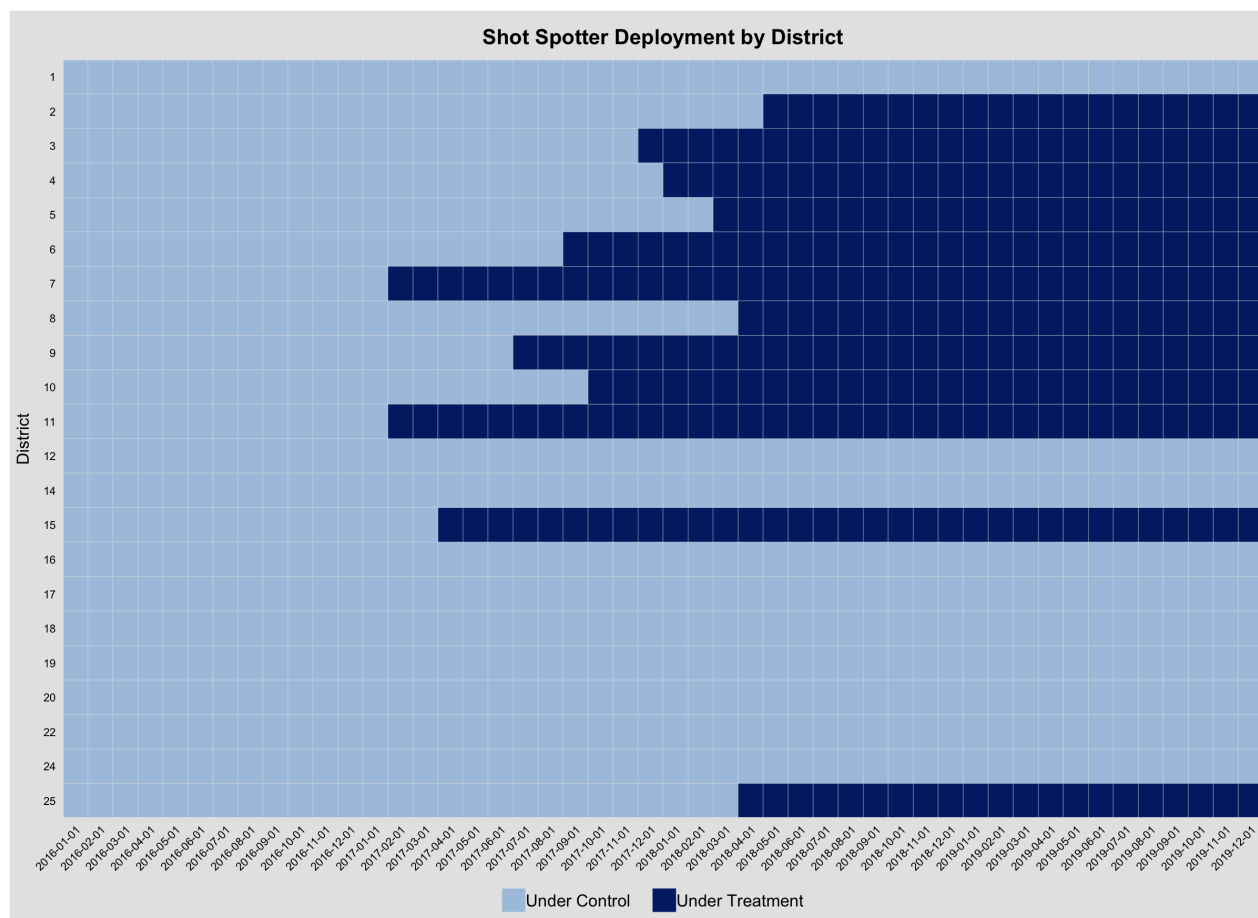


Figure 2. ShotSpotter deployment schedule by CPD District and Year-Month.

Treated district-periods are depicted in dark blue, untreated in light blue.

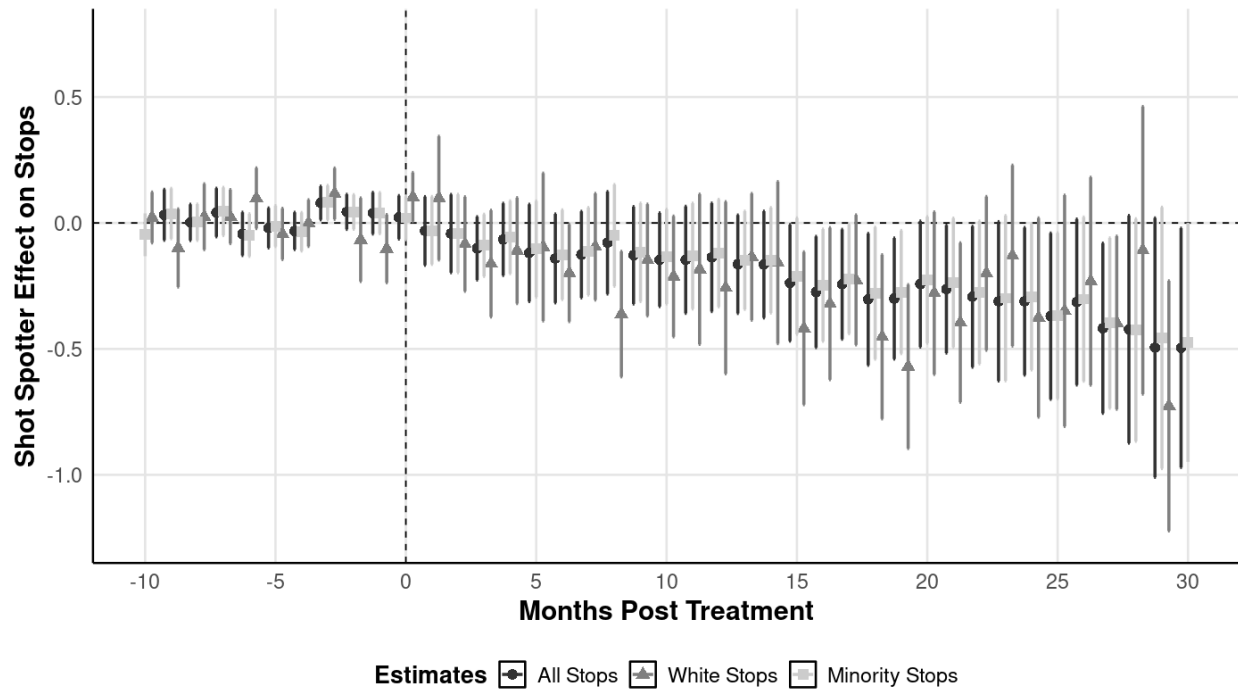


Figure 3. Dynamic estimates of relationship between ShotSpotter deployment in a district and the volume of stops reported by CPD.

Average treatment effects on the treated (points) for the effect of ShotSpotter on each outcome are depicted with 95% confidence intervals (lines) before and after ShotSpotter is deployed in a district. Dynamic treatment effects estimation is performed using Matrix Completion as implemented in the R package ‘fect’ (Athey et al. 2021); inference is based on a block bootstrap with 1,000 iterations.

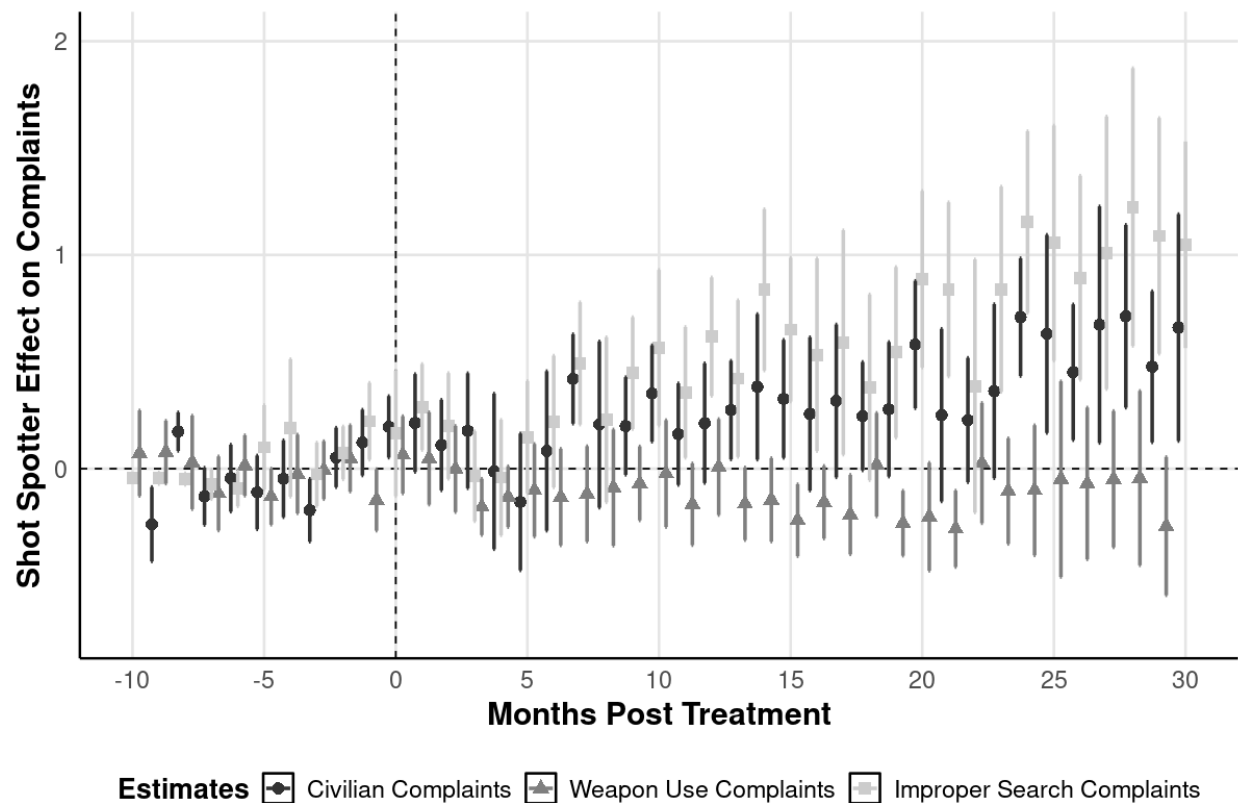


Figure 4. Dynamic estimates of relationship between ShotSpotter deployment in a district and civilian complaints.

Average treatment effects on the treated (points) for the effect of ShotSpotter on each outcome are depicted with 95% confidence intervals (lines) before and after ShotSpotter is deployed in a district. Dynamic treatment effects estimation is performed using Matrix Completion as implemented in the R package ‘fect’ (Athey et al. 2021); inference is based on a block bootstrap with 1,000 iterations.

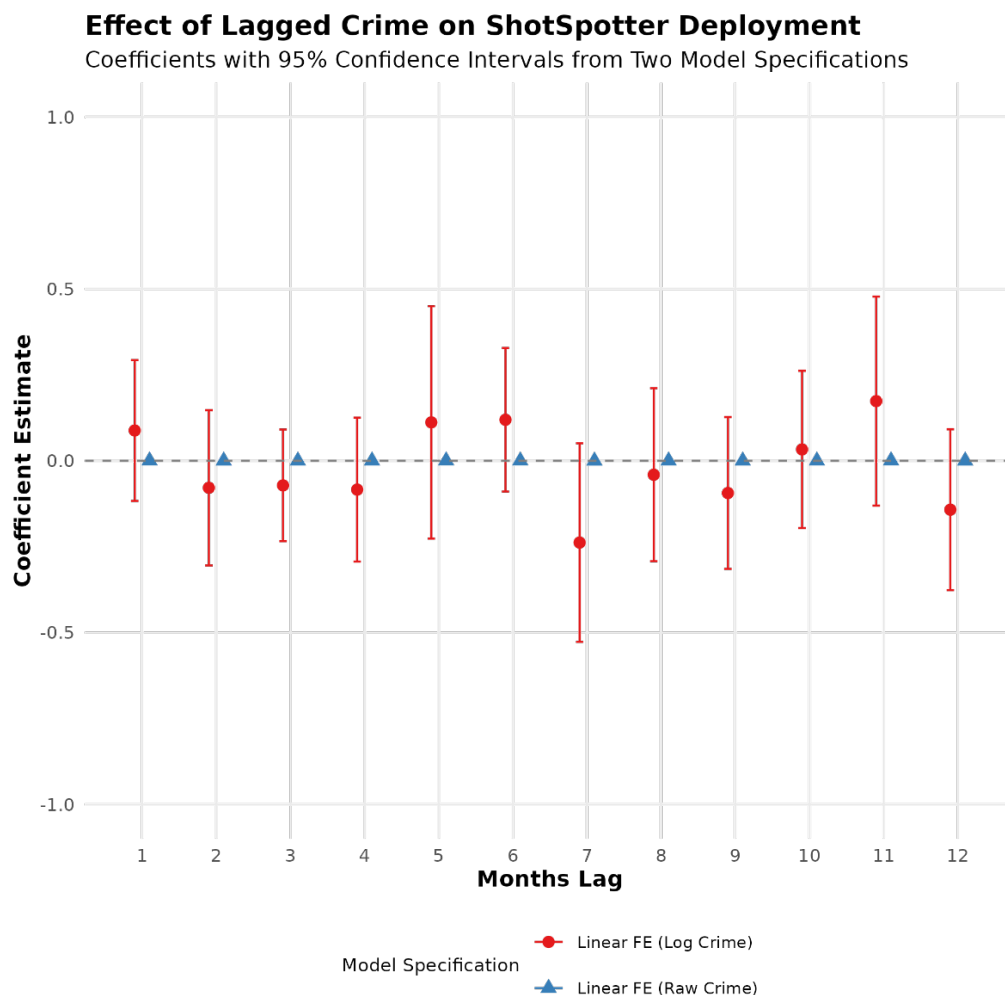


Figure 5. Crime trajectory does not predict deployment of ShotSpotter.

We test whether recent historical changes in crime within a district are meaningfully associated with the arrival of ShotSpotter. We conduct this test with a pair of linear probability models (LPMs), wherein we regress the indicator of ShotSpotter deployment onto lagged monthly measures of crime (either in levels or logs) in the district up to one year in advance. Regardless of whether we consider raw logged measures of crime, we observe no evidence that changes in a district's monthly crime over the prior year are predictive of ShotSpotter deployment.

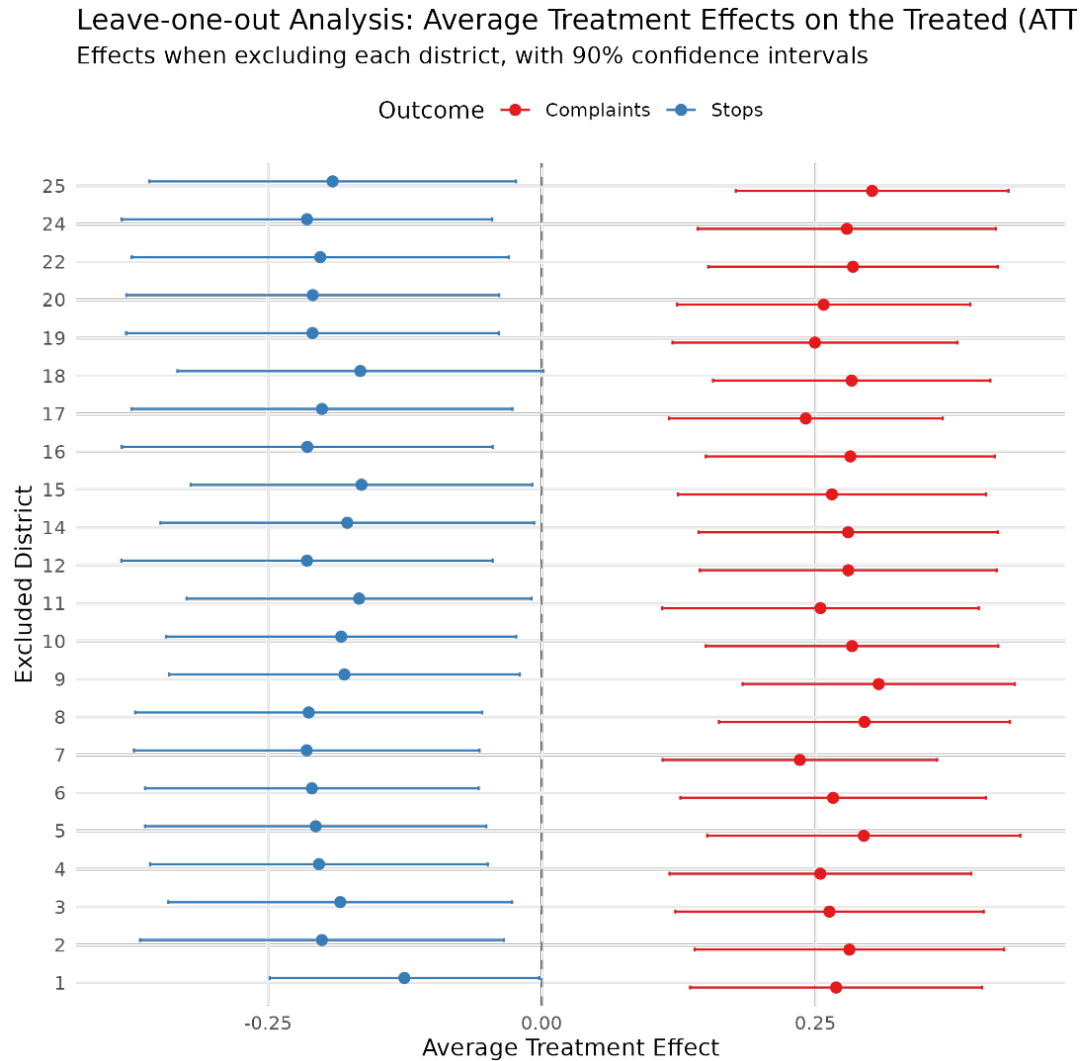


Figure 6. Leave-one-out Analysis: Effects of GDT are general, not driven by any one district.

We conduct a leave-one-out analysis by repeatedly estimating our two key models (effects on stops and complaints) while systematically omitting each individual police district from the sample. Point estimates and 90% confidence intervals are plotted for the effect on stops (blue) and complaints (red). This exercise gauges the sensitivity of our estimates to any single district's exclusion (vertical axis). This analysis helps to ensure that our findings are broad and relate to the wider set of treated districts, and thus that they may be trusted to generalize to new jurisdictions.

TABLES

Table 1. Descriptive Statistics

	Mean	SD	Min	Max	N
ShotSpotter	0.30	0.46	0.00	1.00	1056
Log(Stops)	6.01	0.60	4.13	7.40	1056
Log(White Stops)	3.32	0.98	0.00	5.37	1056
Log(Minority Stops)	5.88	0.68	3.97	7.38	1056
Log(Traffic Stops)	4.79	3.24	0.00	8.83	1056
Log(Gun Discharge Arrests)	0.38	0.48	0.00	1.95	1056
Log(Unlawful Use of Weapon Arrests)	2.23	0.96	0.00	4.13	1056
Log(Crimes)	6.87	0.33	5.71	7.52	1056
Crime Closure Rate	0.19	0.07	0.09	0.47	1056
Log(District Staffing)	5.72	0.17	5.31	6.08	1056
Log(Complaints)	1.34	0.68	0.00	3.18	1056
Log(Weapon Complaints)	0.16	0.32	0.00	1.39	1056
Log(Improper Search Complaints)	0.51	0.69	0.00	2.83	1056
Log(Gun Crimes)	3.74	0.80	1.39	5.17	1056

Notes: Data covers observations in Chicago from 2016 through 2019. Observations are at the district by month-year level, yielding 1,056 observations in this estimation sample.

Table 2. Effect of ShotSpotter on Investigatory Stops

	Stops	White Stops	Minority Stops	District Staffing
ShotSpotter	-0.194** (0.095) [-0.379, -0.008]	-0.234** (0.107) [-0.443, -0.025]	-0.177* (0.095) [-0.364, 0.010]	-0.002 (0.013) [-0.027, 0.023]
Observations	1,056	1,056	1,056	1,056

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$; Outcome variables are log-transformed; Estimation is performed using Matrix Completion as implemented in the R package ‘fect’ (Athey et al. 2021); Standard errors are based on a block bootstrap with 1,000 iterations. 95% confidence intervals are provided below estimates in brackets.

Table 3. Average Effect of ShotSpotter on Crime and Arrests

	Crimes	Gun Crimes	Crime Closure Rate	Gun Discharge Arrests	Unlawful Use of Weapons Arrests
ShotSpotter	-0.028 (0.021) [-0.070, 0.014]	0.077 (0.047) [-0.015, 0.169]	0.013* (0.008) [-0.002, 0.029]	0.105** (0.046) [0.016, 0.194]	0.128* (0.078) [-0.024, 0.280]
Observations	1,056	1,056	1, 056	1,056	1,056

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$; Outcome variables are log-transformed with the exception of Crime Closure Rate; Estimation is performed using Matrix Completion as implemented in the R package ‘fect’ (Athey et al. 2021); Standard errors are based on a block bootstrap with 1,000 iterations. 95% confidence intervals are provided below estimates in brackets.

Table 4. Average Effect of ShotSpotter on Civilian Complaints.

	Complaints	Weapon Complaints	Improper Search Complaints
ShotSpotter	0.275*** (0.082) [0.114, 0.436]	-0.108** (0.045) [-0.196, -0.019]	0.521*** (0.108) [0.310, 0.733]
Observations	1,056	1,056	1,056

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$; Outcome variables are log-transformed; Estimation is performed using Matrix Completion as implemented in the R package ‘fect’ (Athey et al. 2021); Standard errors are based on a block bootstrap with 1,000 iterations. 95% confidence intervals are provided below estimates in brackets.

STATISTICAL APPENDIX

Data Description

Data records covering the CPD's Investigatory Stop Report (ISRs) were collected from the CPD's online database, available at <https://home.chicagopolice.org/statistics-data/isr-data/> (Accessed 5/28/2024). This database contains observations at the level of the individual stop and provides time, location, and stopped-suspect characteristics. For our study, this data is aggregated to the district-month level for all months between January 2016 and December 2019. This yields 1,056 total observations from 22 unique districts over 48 time periods. District staffing data was acquired by FOIA request (FOIA P967257) and aggregated to the district-month level. Traffic stop data was obtained from a public repository published by WBEZ News, available at <https://interactive.wbez.org/traffic-stops/chicago-police/> (Accessed 6/14/2024).

Crime report data was collected from City of Chicago's online data portal, available at https://data.cityofchicago.org/Public-Safety/Crimes-2001-to-Present/ijzp-q8t2/about_data (Accessed 5/28/2024). This database contains observations at the level of the crime, and provides time, location, and a Boolean arrest flag that measures whether the given crime was resolved through an arrest. This crime data was aggregated and merged at the district-month level. Data on firearm arrests was acquired via FOIA request filed with the CPD (FOIA P949644). This dataset provides observations at the level of the charge indexed by arrests. Arrests related to weapons and firearms charges were aggregated and merged by district-month. Complaints data was acquired from the City of Chicago's online data portal, available at https://data.cityofchicago.org/Public-Safety/COPA-Cases-Summary/mft5-nfa8/about_data (Accessed 6/6/2024). This data contains observations at the level of the complaint and provides date, location (CPD beat), and complaint characteristics. Civilian's complaints were aggregated to the district-month level and merged with our estimation sample.

Finally, ShotSpotter alert data was acquired from City of Chicago's online data portal, available at https://data.cityofchicago.org/Public-Safety/Violence-Reduction-Shotspotter-Alerts/3h7q-7mdb/about_data (Accessed 6/5/2024). This data contains observation at the level of the alert generated by ShotSpotter, and provides date, location, and incident characteristics. These alerts were aggregated and merged by district-month level. Finally, ShotSpotter implementation dates by district were obtained from

Connealy et al. (Connealy et al. 2024). This rollout schedule was merged with our data by district and month.

Dependent Variable Definitions

Stops_{it}: Variable counting the number of stops in district i during month-year period t . This outcome is log transformed using the natural logarithm of the stop count plus one, $\ln(\text{Stops}_{it}+1)$.

White Stops_{it}: Variable counting the number of stops of white suspects in district i during month-year period t . White suspects are counted when the variable *RACE_CODE_CD* is equal to “WHT.” This outcome is log transformed using the natural logarithm of the stop count plus one, $\ln(\text{White Stops}_{it}+1)$.

Minority Stops_{it}: Variable counting the number of stops of minority suspects in district i during month-year period t . Minority stops are measured by subtracting White Stops_{it} from Stops_{it} during a district month-year. This outcome is log transformed using the natural logarithm of the stop count plus one, $\ln(\text{Minority Stops}_{it}+1)$.

Traffic Stops_{it}: Variable counting the total number of traffic stops in district i during month-year period t . This outcome is log transformed using the natural logarithm of the stop count plus one, $\ln(\text{Traffic Stops}_{it}+1)$.

Staffing_{it}: Variable measuring the staffing headcount in each police district i during month-year period t . This variable measures only non-recruit training assignments (Assigned unit not equal to “044”). Personnel are counted in a detail unit if provided, otherwise they are counted in their assigned unit. To capture redeployment within a month, we measure the fraction of a month that each person is assigned to a given unit and sum headcounts by police district for each month-year. For example, if Person A spent 15 days in Unit 001 and 15 days in Unit 002 in a month with 30 days, they would be counted as 0.5 for District 1 and 0.5 for District 2. This measure is log transformed using the natural logarithm of the staffing level, $\ln(\text{Staffing}_{it}+1)$.

Crimes_{it}: Variable counting the total number of crimes reported in district i during month-year period t . This outcome is log transformed using the natural logarithm of the crime count plus one, $\ln(\text{Crimes}_{it}+1)$.

Gun Crimes_{it}: Variable counting the total number of crimes reported in district i during month-year period t that include “firearm” or “handgun” in the crime description. This outcome is log transformed using the natural logarithm of the gun crime count plus one, $\ln(\text{Gun Crimes}_{it}+1)$.

Crime Closure Rate_{it}: Variable measuring the arrest rate of crimes reported in district i during month-year period t . This outcome measures the proportion of crimes with the variable *Arrest* equal to “TRUE” to the total number of crimes reported in a district during the time period.

Gun Discharge Arrests_{it}: Variable counting the total number of gun discharge arrests in district i during month-year period t . This outcome counts the number of arrests with FBI Code “15” and “DISCH” contained in the statute description, and is log transformed using the natural logarithm of the arrest count plus one, $\ln(\text{Gun Discharge Arrests}_{it} + 1)$.

Unlawful Use of Weapon Arrests_{it}: Variable counting the total number of unlawful use of weapon arrests in district i during month-year period t . This outcome counts the number of arrests with FBI Code “15” and “UW” contained in the statute description, and is log transformed using the natural logarithm of the arrest count plus one, $\ln(\text{Unlawful Use of Weapon Arrests}_{it} + 1)$.

Complaints_{it}: Variable counting the total number of civilian complaints in district i during month-year time period t . This outcome counts the number of complaints initiated by civilians, complaints with *ASSIGNMENT* not equal to “BIA” and *CASE_TYPE* equal to “Complaint”. This outcome is log transformed using the natural logarithm of the complaint count plus one, $\ln(\text{Complaints}_{it} + 1)$.

Weapon Complaints_{it}: Variable counting the total number of civilian complaints involving weapons and firearms in district i during month-year period t . This outcome counts complaints with *CURRENT_CATEGORY* containing “weapon” or “firearm” (case insensitive), with *ASSIGNMENT* not equal to “BIA” and *CASE_TYPE* equal to “Complaint”. This outcome is log transformed using the natural logarithm of the complaint count plus one, $\ln(\text{Weapon Complaints}_{it} + 1)$.

Improper Search Complaints_{it}: Variable counting the total number of civilian complaints involving improper search in district i during month-year period t . This outcome counts complaints with *CURRENT_CATEGORY* containing “improper search” (case insensitive), with *ASSIGNMENT* not equal to “BIA” and *CASE_TYPE* equal to “Complaint”. This outcome is log transformed using the natural logarithm of the complaint count plus one, $\ln(\text{Improper Search Complaints}_{it} + 1)$.

ShotSpotter Calls_{it}: Variable counting the total number of emergency calls initiated by ShotSpotter district i

during month-year period t . This outcome is log transformed using the natural logarithm of the call count plus one, $\ln(\text{ShotSpotter Calls}_{it}+1)$.

Rounds Fired_{it}: Variable counting the total number of firearm discharge rounds detected by ShotSpotter in district i during month-year period t . This outcome is log transformed using the natural logarithm of the rounds detected plus one, $\ln(\text{Rounds Fired}_{it}+1)$.

Independent Variable Definitions

ShotSpotter_{it}: A binary (0/1) variable that measures whether ShotSpotter has been deployed and become operational in a district i during month-year period t . This measure is taken from Connealy et al (Connealy et al. 2024), and is equal to 1 in the first month after ShotSpotter becomes operational in a district.

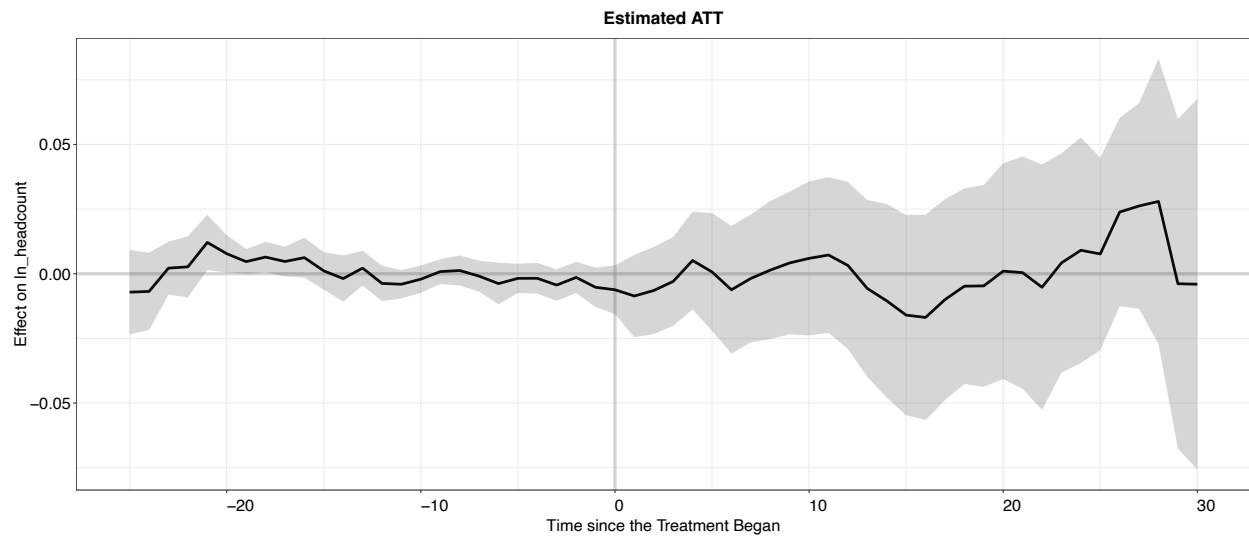


Figure S1. Dynamic estimates of relationship between ShotSpotter deployment in a district and district headcount (Matrix Completion).

We observe no evidence of systemic change in police district headcounts in the months following ShotSpotter deployment in a police district. Dynamic treatment effects estimation is performed using Matrix Completion as implemented in the R package ‘fect’ (Athey et al. 2021); 95% confidence intervals are provided for each period, based on a block bootstrap with 1,000 iterations.

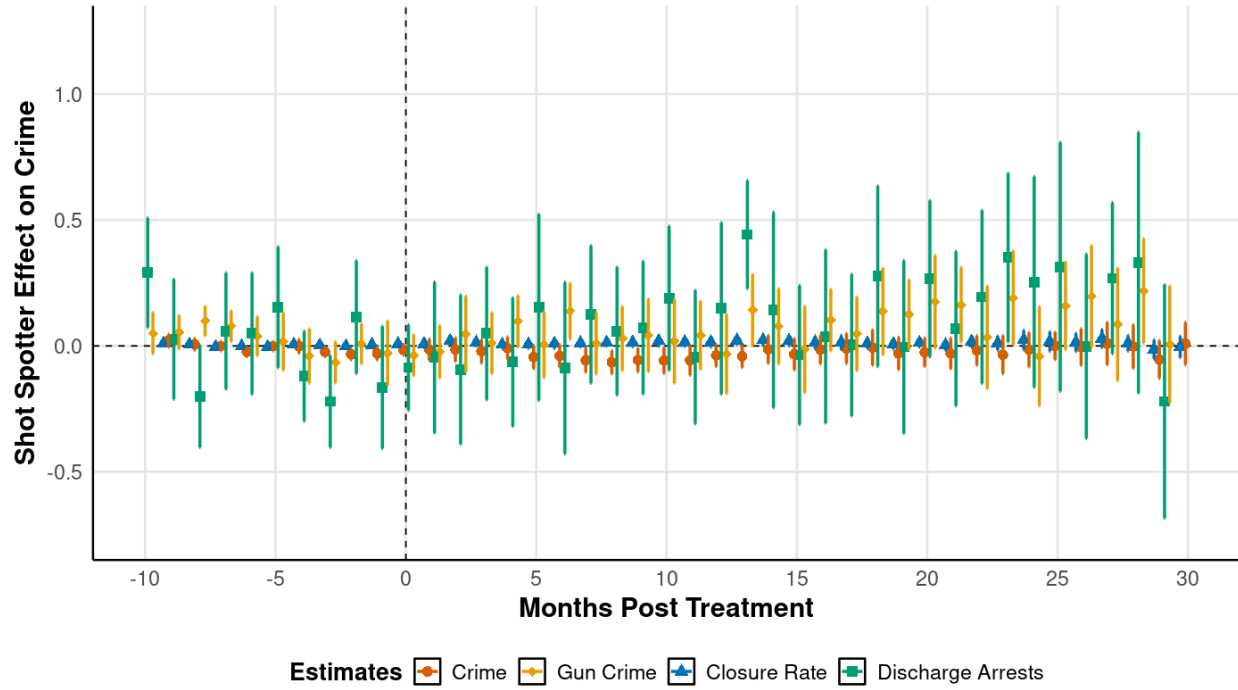


Figure S2. Dynamic estimates of relationship between ShotSpotter deployment in a district and crime, closure rate, and firearm discharge arrests (Matrix Completion).

We observe a systematic increase on discharge arrests and a small increase in the closure rate of crime, with no apparent increase in the crime rate. Dynamic treatment effects estimation is performed using Matrix Completion as implemented in the R package ‘fect’ (Athey et al. 2021); 95% confidence intervals are provided for each period, based on a block bootstrap with 1,000 iterations.

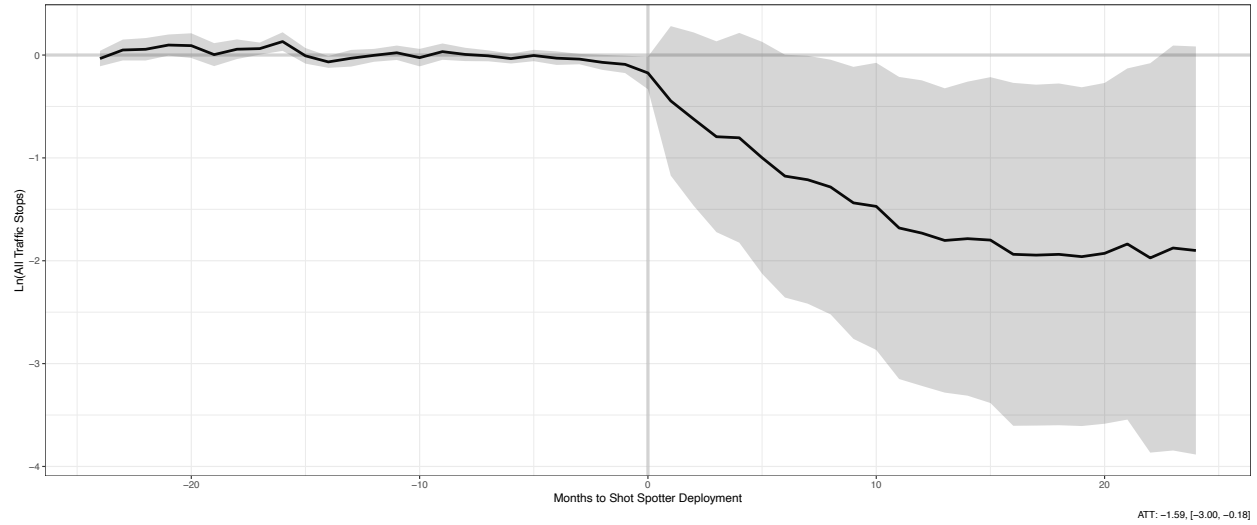


Figure S3. Dynamic estimates of relationship between ShotSpotter deployment in a district and traffic stops (Matrix Completion).

The average treatment effects on the treated (bold line) for the effect of ShotSpotter on traffic stops are depicted with 95% confidence intervals (shaded region) before and after ShotSpotter is deployed in a district. Dynamic treatment effects estimation is performed using Matrix Completion as implemented in the R package ‘fect’ (Athey et al. 2021); inference is based on a block bootstrap with 1,000 iterations.

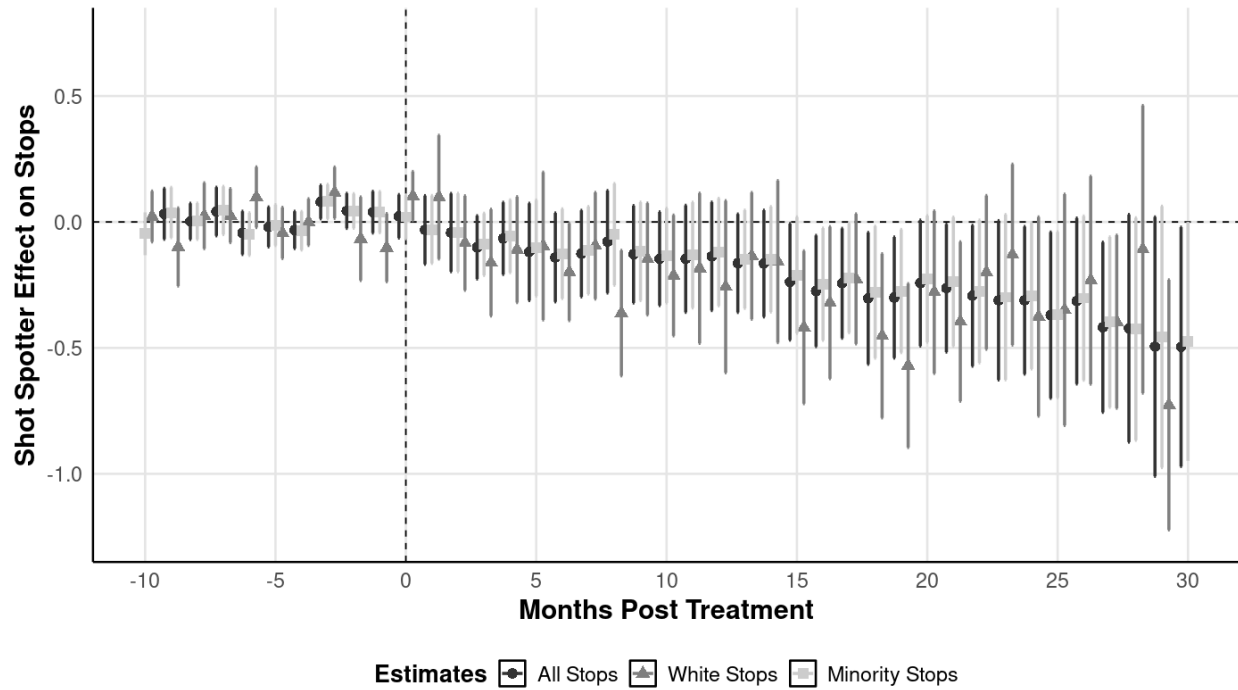


Figure S4. Dynamic estimates of relationship between ShotSpotter deployment in a district and the volume of stops reported by CPD (DID2S).

Average treatment effects on the treated (points) for the effect of ShotSpotter on each outcome are depicted with 95% confidence intervals (lines) before and after ShotSpotter is deployed in a district. Dynamic treatment effects estimation is performed two-stage DID (Gardner 2022); standard errors are clustered by precinct.

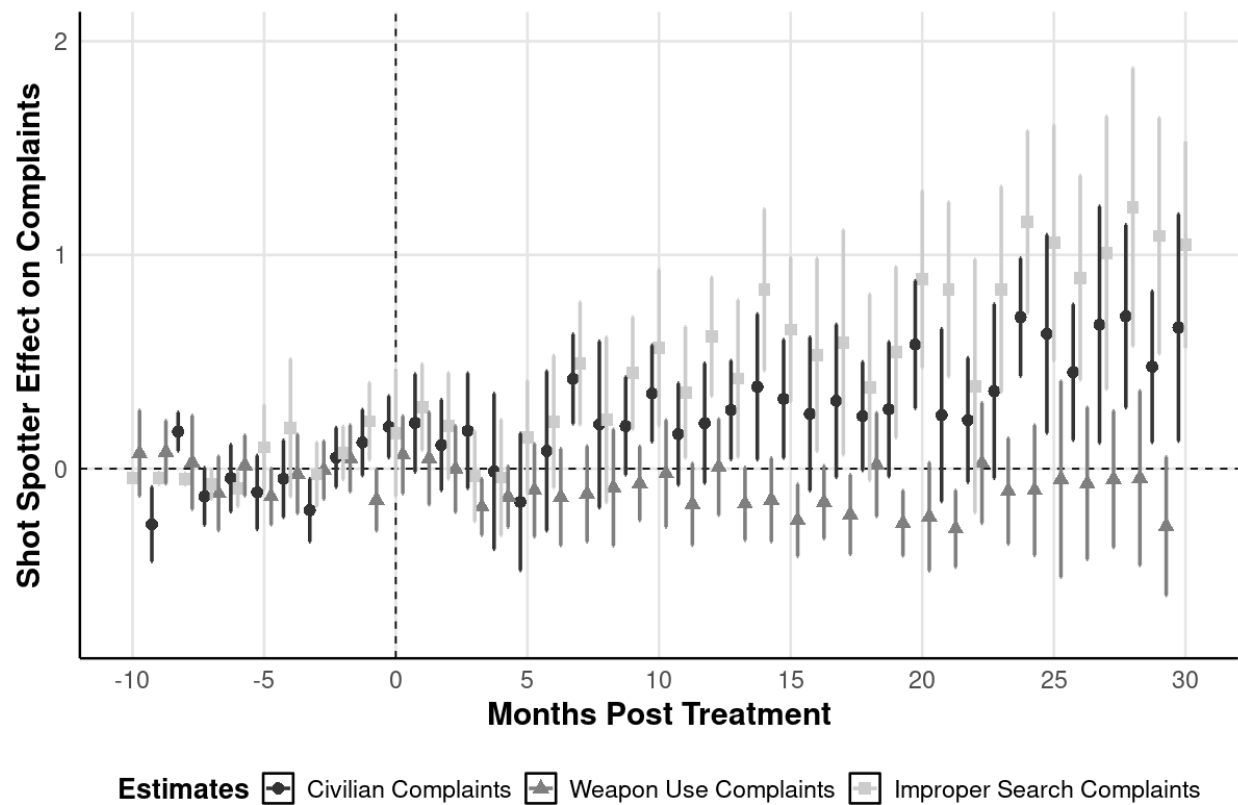


Figure S5. Dynamic estimates of relationship between ShotSpotter deployment in a district and civilian complaints (DID2S).

Average treatment effects on the treated (points) for the effect of ShotSpotter on each outcome are depicted with 95% confidence intervals (lines) before and after ShotSpotter is deployed in a district. Dynamic treatment effects estimation is performed using two-stage DID (Gardner 2022); standard errors are clustered by precinct.

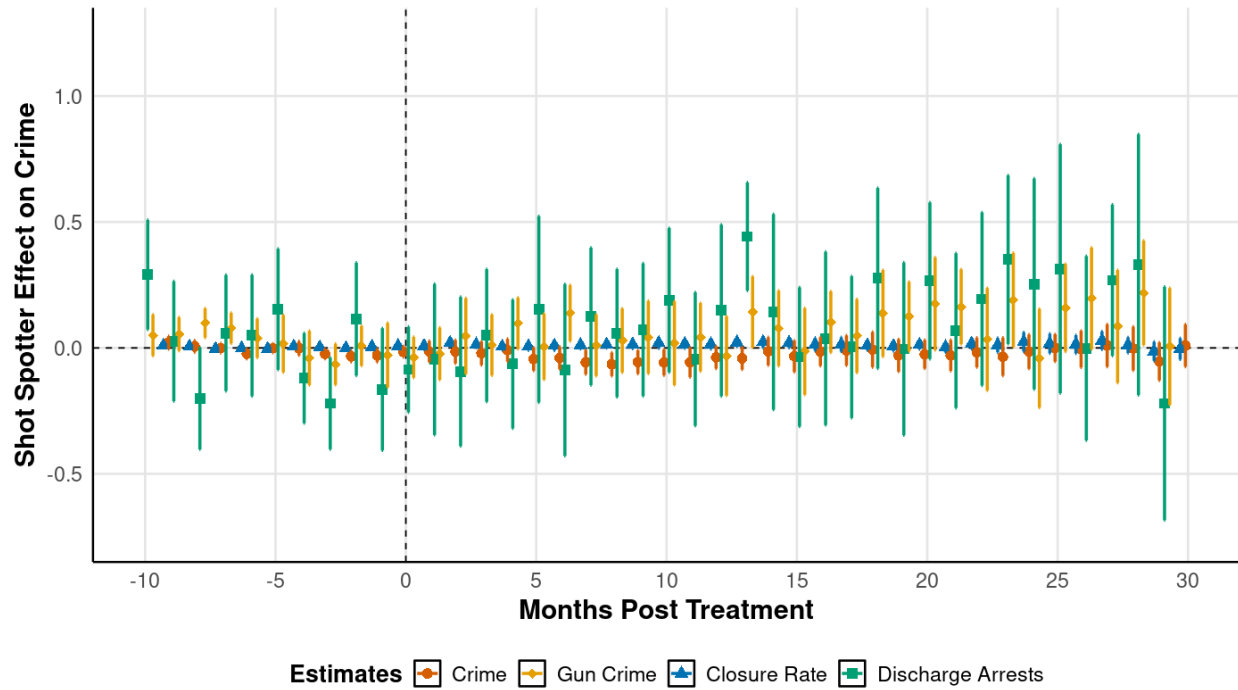


Figure S6. Dynamic estimates of relationship between ShotSpotter deployment in a district and crime, closure rate, and firearm discharge arrests (DID2S)

We observe a systematic increase on discharge arrests and a small increase in the closure rate of crime, with no apparent increase in the crime rate. Dynamic treatment effects estimation is performed using two-stage DID (Gardner 2022); standard errors are clustered by precinct..

Table S1. Average Effect of ShotSpotter Deployment on ShotSpotter Initiated Calls (Matrix Completion)

DV	ShotSpotter Calls	Rounds Fired
ShotSpotter	3.968*** (0.194) [3.588, 4.348]	4.937*** (0.279) [4.391, 5.484]
Observations	1,056	1,056

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$; Outcome variables are log-transformed; Estimation is performed using Matrix Completion as implemented in the R package ‘fect’ (Athey et al. 2021); Standard errors are based on a block bootstrap with 200 iterations. 95% confidence intervals are provided below estimates in brackets.

Table S2. Average Effect of ShotSpotter on Investigatory Stops (DID2S)

	Stops	White Stops	Minority Stops	District Staffing
ShotSpotter	-0.207** (0.100) [-0.404, -0.010]	-0.242** (0.109) [-0.456, -0.029]	-0.193* (0.100) [-0.390, 0.005]	-0.001 (0.012) [-0.025, 0.022]
Observations	1,056	1,056	1,056	1,056

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$; Outcome variables are log-transformed; Estimation is performed using two-stage difference-in-differences (Gardner 2022); standard errors are clustered by precinct. 95% confidence intervals are provided in brackets.

Table S3. Average Effect of ShotSpotter on Crime and Arrests (DID2S)

	Crimes	Gun Crimes	Crime Closure Rate	Gun Discharge Arrests	Unlawful Use of Weapon Arrests
ShotSpotter	-0.031 (0.023) [-0.077, 0.014]	0.072 (0.047) [-0.020, 0.164]	0.015* (0.008) [-0.002, 0.031]	0.095** (0.045) [0.006, 0.184]	0.133* (0.076) [-0.016, 0.281]
Observations	1,056	1,056	1,056	1,056	1,056

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$; Outcome variables are log-transformed with the exception of Crime Closure Rate; Estimation is performed using two-stage DID (Gardner 2022); standard errors are clustered by district. 95% confidence intervals are provided in brackets.

Table S4. Average Effect of ShotSpotter on Civilian Complaints (DID2S)

	Complaints	Weapon Complaints	Improper Search Complaints
ShotSpotter	0.284*** (0.083) [0.120, 0.448]	-0.124** (0.051) [-0.223, -0.025]	0.546*** (0.115) [0.320, 0.772]
Observations	1,056	1,056	1,056

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$; Outcome variables are log-transformed; Estimation is performed using two-stage DID (Gardner 2022); standard errors are clustered by district. 95% confidence intervals are provided in brackets.